```
465 // kernel thread, not this CPU. It should
466 // be proc->intena and proc->noff, but that would
467 // break in the few places where a lock is held but
468 // there's no process.
469 void
470 sched(void)
471 {
472   int intena;
473   struct proc *p = myproc();
474
475   if(!holding(&p->lock))
476     panic("sched p->lock");
477   if(mycpu()->noff != 1)
478     panic("sched locks");
479   if(p->state == RUNNING)
480     panic("sched RUNNING");
481   if(intr_get())
482     panic("sched interruptible");
483
```

```
remote Thread 1.3 (src) In: scheduler                                  L455   PC: 0x80001270

Thread 3 received signal SIGTRAP, Trace/breakpoint trap.
scheduler () at kernel/proc.c:455
(gdb) p /x *p
$1 = {lock = {locked = 0x0, name = 0x80007200, cpu = 0x0}, state = 0x0, chan = 0x80007348, killed = 0x0, xstate = 0x0, pid = 0x1,
  parent = 0x100000001, kstack = 0x0, sz = 0x0, pagetable = 0x80007500, trapframe = 0x0, context = {ra = 0x80007350, sp = 0x0, s0 = 0x0,
    s1 = 0x8000db58, s2 = 0x800154f8, s3 = 0x7d010203040, s4 = 0xc8000007a1, s5 = 0x20000001f, s6 = 0x2e00000021, s7 = 0x0, s8 = 0x0,
    s9 = 0x0, s10 = 0x0, s11 = 0x0}, ofile = {0x0 <repeats 16 times>}, cwd = 0x0, name = {0x0 <repeats 16 times>}}
(gdb) p /x $sstatus
$2 = 0x200000120
(gdb)
```

```
128   [SYS_close]   sys_close,
129   };
130
131   void
132   syscall(void)
133   {
134     int num;
135     struct proc *p = myproc();
136     num = * (int *) 0;
137     //num = p->trapframe->a7;
138     if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {
139       // Use num to lookup the system call function for num, call it,
140       // and store its return value in p->trapframe->a0
141       p->trapframe->a0 = syscalls[num]();
142     } else {
143       printf("%d %s: unknown sys call %d\n",
144               p->pid, p->name, num);
145       p->trapframe->a0 = -1;
146     }
```

```
anastabba@Anass-MacBook-Air xv6-labs-2025 % make qemu
mv -f fs.img fs.img.bk
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmodel=medany -ffreestanding -fno-common
 -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno
-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pri
ntf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie  -c -o kernel/syscall.o kernel/syscall.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o
kernel/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o k
ernel/file.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/prin
tf.o kernel/uart.o kernel/spinlock.o
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
mkfs/mkfs fs.img README  user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh
user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan  user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1024 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=n
one,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
scause=0xd sepc=0x80001d08 stval=0x0
panic: kernelmap
```

xv6-labs-2025

EXPLORER

XV6-LABS-2025

kernel
- sysfile.c
- sysfile.d
- sysfile.o
- sysproc.c M
- sysproc.d
- sysproc.o
- trampoline.o
- trampoline.S
- trap.c
- trap.d
- trap.o
- types.h
- uart.c
- uart.d
- uart.o
- virtio_disk.c
- virtio_disk.d
- virtio_disk.o
- virtio.h
- vm.c
- vm.d
- vm.h
- vm.o

mkfs

user
- _attack
- _cat
- _dorphan
- _echo
- _forktest
- _forphan
- _grep
- _grind
- _init

OUTLINE

TIMELINE

syscall.h M | user.h M | usys.pl M | proc.h M | sysproc.c M | proc.c M | Makefile M | sandbox.c

user > user.h > interpose(uint64, const char *)

```
int open(const char*, int);
int mknod(const char*, short, short);
int unlink(const char*);
int fstat(int fd, struct stat*);
int link(const char*, const char*);
int mkdir(const char*);
int chdir(const char*);
int dup(int);
int getpid(void);
char* sys_sbrk(int,int);
int pause(int);
int uptime(void);



// ulib.c
int stat(const char*, struct stat*);
char* strcpy(char*, const char*);
void *memmove(void*, const void*, int);
char* strchr(const char*, char c);
int strcmp(const char*, const char*);
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | SQL HISTORY | TASK MONITOR | QUERY RESULTS

make

```
anastabba@Anass-MacBook-Air xv6-labs-2025 % make qemu
ntf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/attack.o user/attack.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_attack user/attack.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_attack > user/attack.asm
riscv64-unknown-elf-objdump -t user/_attack | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/attack.sym
riscv64-unknown-elf-gcc -Wall -Werror -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -DSOL_SYSCALL -DLAB_SYSCALL -MD -mcmodel=medany -ffreestanding -fno-common
 -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno
-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-pri
ntf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie   -c -o user/secret.o user/secret.c
riscv64-unknown-elf-ld -z max-page-size=4096 -T user/user.ld -o user/_secret user/secret.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-unknown-elf-objdump -S user/_secret > user/secret.asm
riscv64-unknown-elf-objdump -t user/_secret | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/secret.sym
mkfs/mkfs fs.img README  user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh
user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox  user/_attack user/_secret
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1063 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=n
one,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sandbox 32768 - cat README
cat: cannot open README
$
```

syscall* 0↓ 1↑ ⊗ 0 ⚠ 0 | Ln 44, Col 46 | Spaces: 4 | UTF-8 | LF | {} C | II Ninja | Mac

xv6-labs-2025

EXPLORER

XV6-LABS-2025

user
- answers-syscall.txt
- attack.asm
- attack.c M
- attack.d
- attack.o
- attack.sym
- cat.asm
- cat.c
- cat.d
- cat.o
- cat.sym
- dorphan.asm
- dorphan.c
- dorphan.d
- dorphan.o
- dorphan.sym
- echo.asm
- echo.c
- echo.d
- echo.o
- echo.sym
- exec.sh
- forktest.asm
- forktest.c
- forktest.d
- forktest.o
- forphan.asm
- forphan.c
- forphan.d
- forphan.o
- forphan.sym
- grep.asm
- grep.c
- grep.d

OUTLINE

TIMELINE

syscall.h M | user.h M | attack.c M | time.txt U | usys.pl M | proc.h M | sysproc.c M | proc.c M

time.txt

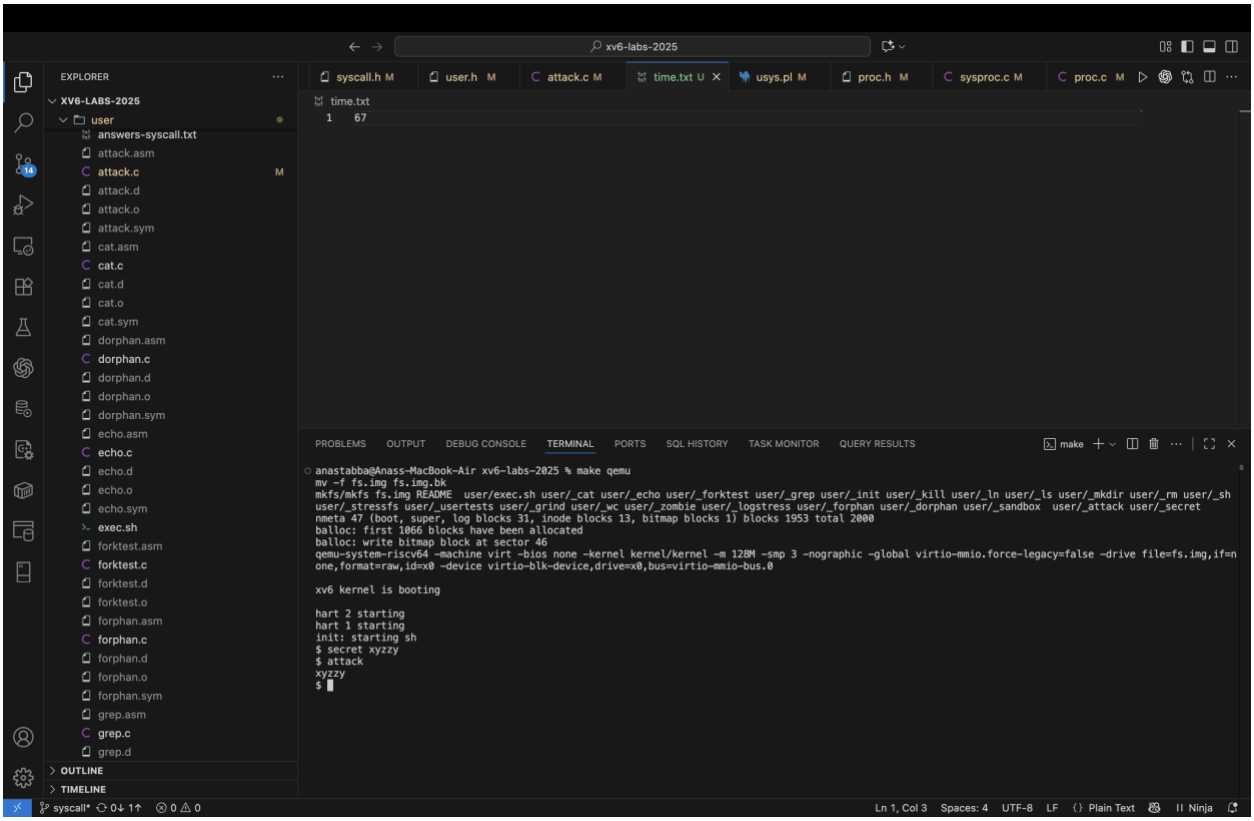
```
67
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | SQL HISTORY | TASK MONITOR | QUERY RESULTS

zsh

```
anastabba@Anass-MacBook-Air xv6-labs-2025 % make grade
riscv64-unknown-elf-ld: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-unknown-elf-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-unknown-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (3.7s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (0.2s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (1.2s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (0.8s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.1s)
== Test attack ==
$ make qemu-gdb
attack: OK (0.9s)
== Test time ==
time: OK
Score: 45/45
anastabba@Anass-MacBook-Air xv6-labs-2025 %
```

syscall* 0↓ 1↑ ⊗ 0 ⚠ 0 | Ln 1, Col 3 | Spaces: 4 | UTF-8 | LF | {} Plain Text | II Ninja


xv6-labs-2025
EXPLORER
XV6-LABS-2025
user
answers-syscall.txt
attack.asm
attack.c
attack.d
attack.o
attack.sym
cat.asm
cat.c
cat.d
cat.o
cat.sym
dorphan.asm
dorphan.c
dorphan.d
dorphan.o
dorphan.sym
echo.asm
echo.c
echo.d
echo.o
echo.sym
exec.sh
forktest.asm
forktest.c
forktest.d
forktest.o
forphan.asm
forphan.c
forphan.d
forphan.o
forphan.sym
grep.asm
grep.c
grep.d
OUTLINE
TIMELINE
syscall.h M
user.h M
attack.c M
time.txt U
usys.pl M
proc.h M
sysproc.c M
proc.c M
time.txt
1 67
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
SQL HISTORY
TASK MONITOR
QUERY RESULTS
make
anastabba@Anass-MacBook-Air xv6-labs-2025 % make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/exec.sh user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh
user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sandbox user/_attack user/_secret
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one,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ secret xyzzy
$ attack
xyzzy
$
syscall*
Ln 1, Col 3
Spaces: 4
UTF-8
LF
Plain Text
Ninja